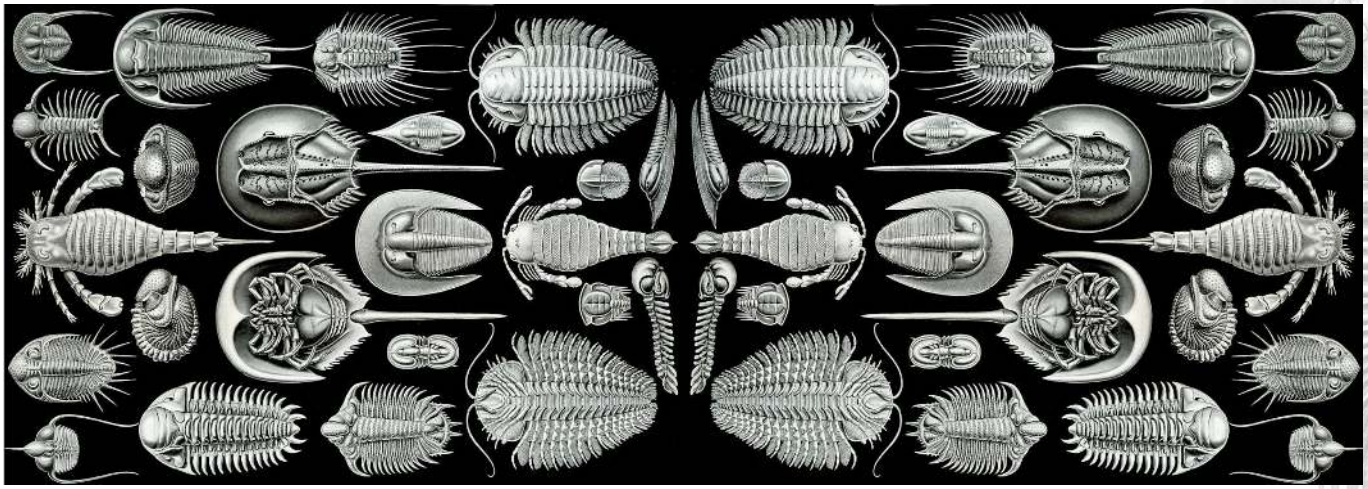
EART27201 Sedimentary Rocks and Fossils



Macrofossils: Trilobites and Echinoderms

Russell Garwood | russell.garwood@manchester.ac.uk

EART27201 Sedimentary Rocks and Fossils



Macrofossils: Trilobites (& Arthropods)

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Introduction and biology

What is a trilobite?

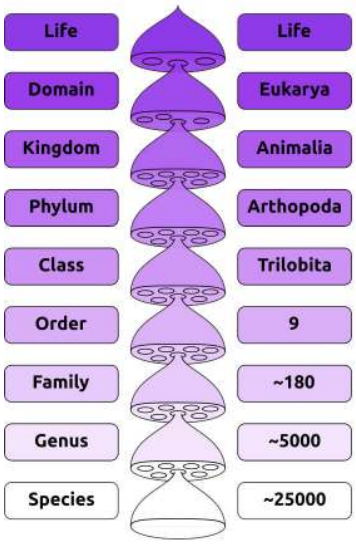


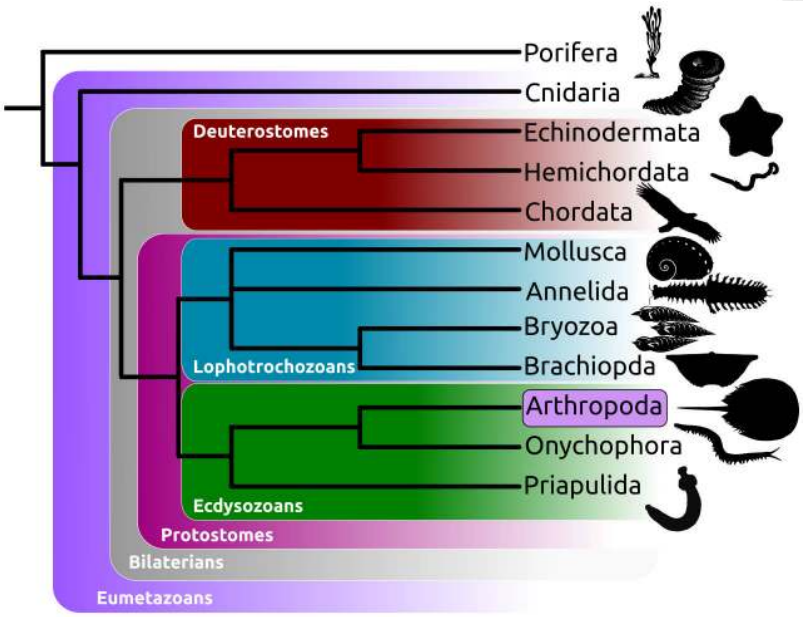




Left: *Neoasaphus cornutus* - The Children's Museum of Indianapolis; Right: *Walliserops trifurcatus* - Kevin Walsh.

Where do they sit on the tree of life?





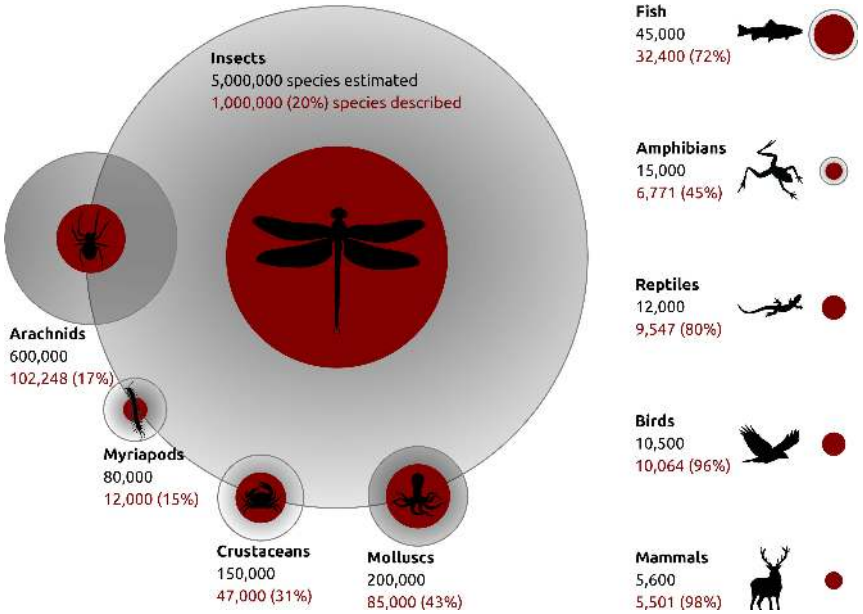
Ecdysozoa | Onychophora | Tardigrada

What is an arthropod?



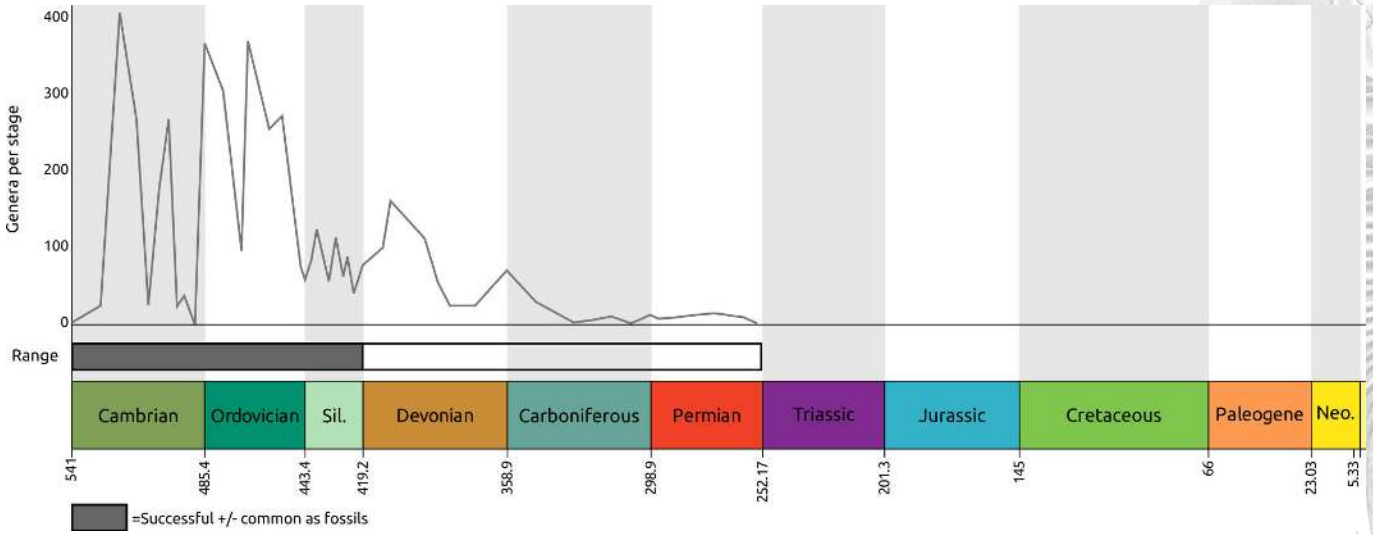
Source: Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons.

Why should I care about arthropods?



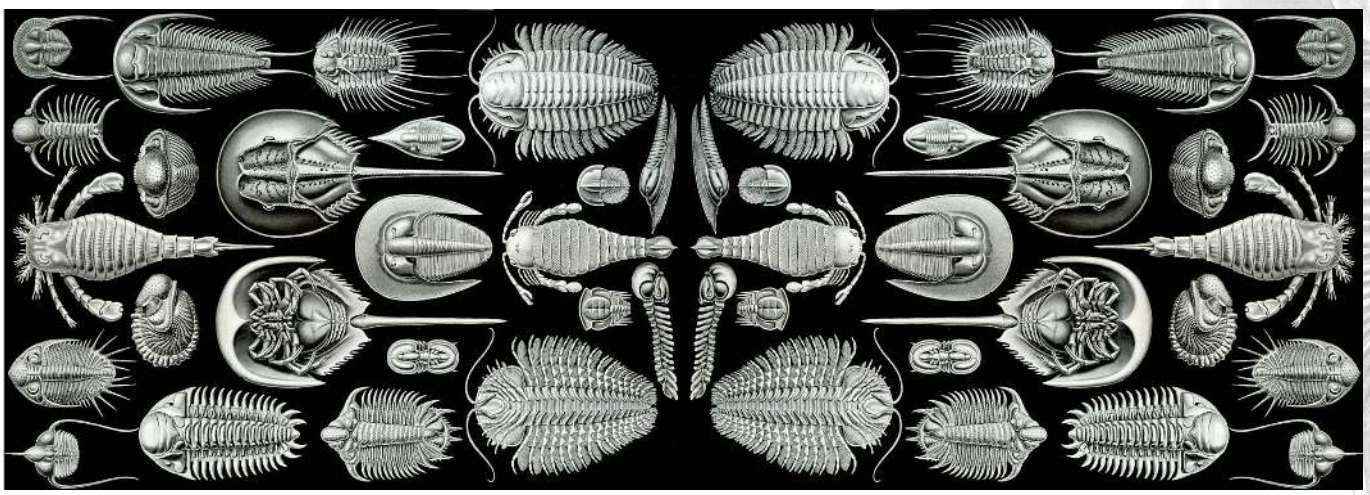
Data from: Chapman, A.D., 2009. Numbers of living species in Australia and the world.
When asked for his view of the Creator, J.B.S. Haldane (1892–1964), famed evolutionary biologist, replied, "God has an inordinate fondness for beetles."

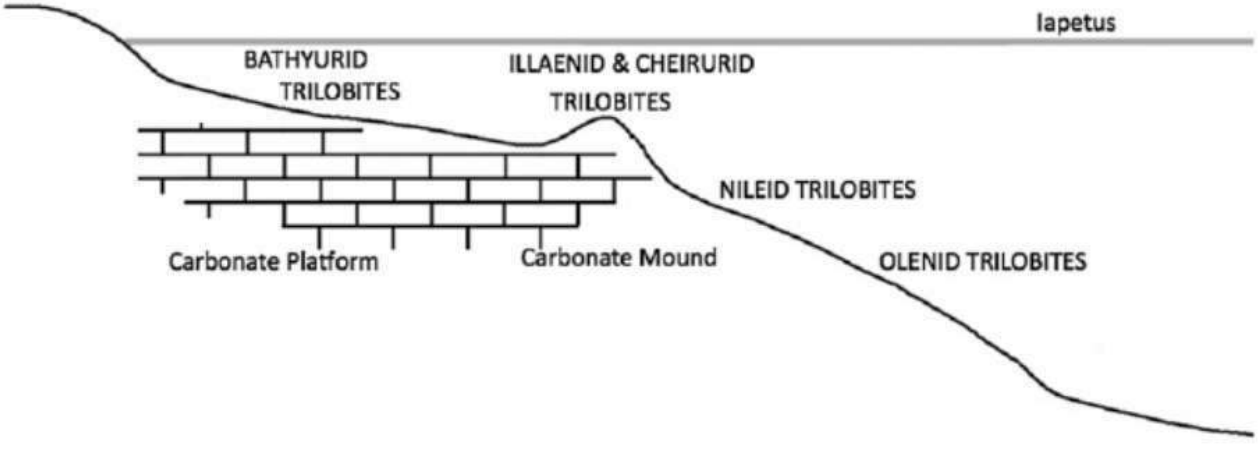
When were the trilobites around?



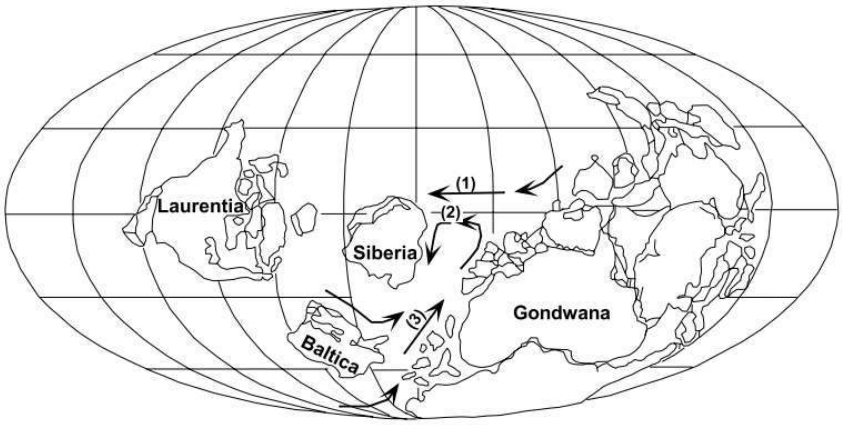
Fossil data from PBDB.

Why should I care about trilobites?





Source: Fortey, R., 2014. The palaeoecology of trilobites. Journal of Zoology, 292(4), 250-259.



(1) north equatorial current

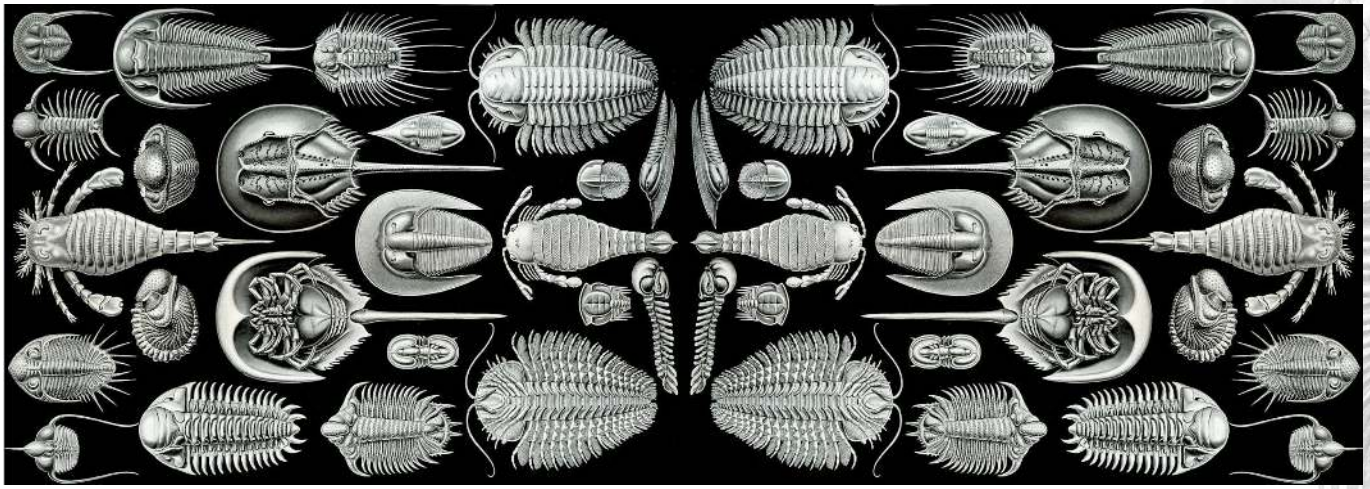
(2) south equatorial current

(3) western Gondwana current

Palaeobiogeography: The scientific study of the geographic distribution of fossils. See also biogeography.
Source: Allaby, M., 2008. Oxford Dictionary of Earth Sciences.

Alvaro, J.J., Elicki, O., Geyer, G., Rushton, A.W. and Shergold, J.H., 2003. Palaeogeographical controls on the Cambrian trilobite immigration and evolutionary patterns reported in the western Gondwana margin. Palaeogeography, Palaeoclimatology, Palaeoecology, 195(1-2), pp.5-35.

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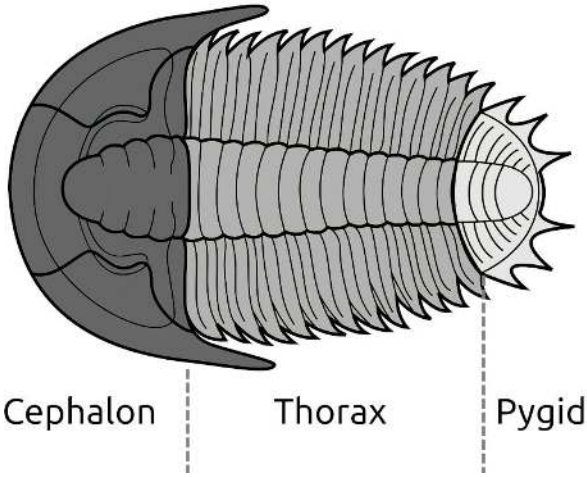


Macrofossils: Trilobites (& Arthropods)

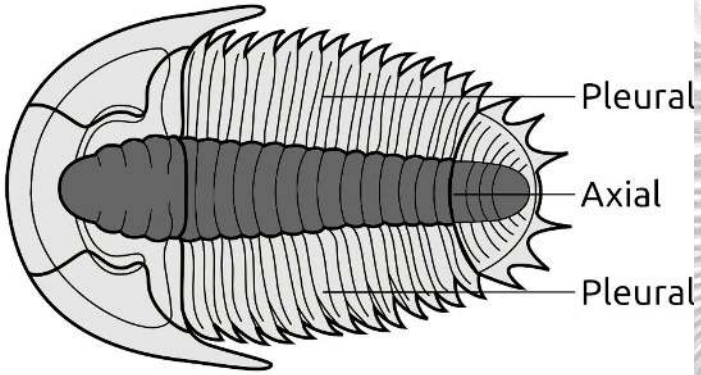
Russell Garwood | russell.garwood@manchester.ac.uk

Morphology / Anatomy

Tagmata



Lobes



All trilobite drawings redrawn after originals by S. M. Gon III. Definitions: Park, C. and Allaby, M. 2013. A dictionary of environment and conservation. Oxford University Press.

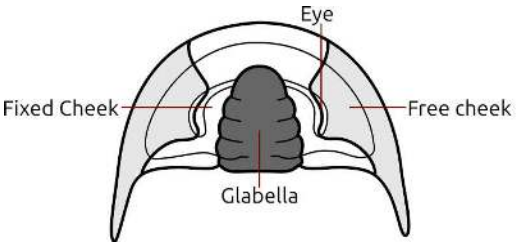
Tagmata: A specialized group of segments in the body of an arthropod.

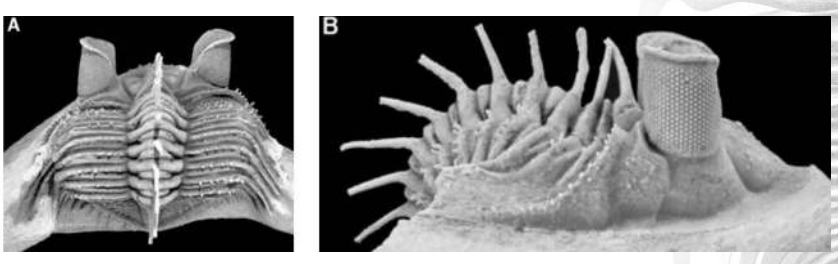
Cephalon: The anterior or head region of a trilobite, which consists of at least five fused segments and is generally semicircular in shape.

Thorax: A mid-body comprising a varying number of segments.

Pygidium: The posterior part of the exoskeleton of a trilobite (Trilobita); it is generally formed by the fusion of several body segments but in some Cambrian forms consisting of a single segment.

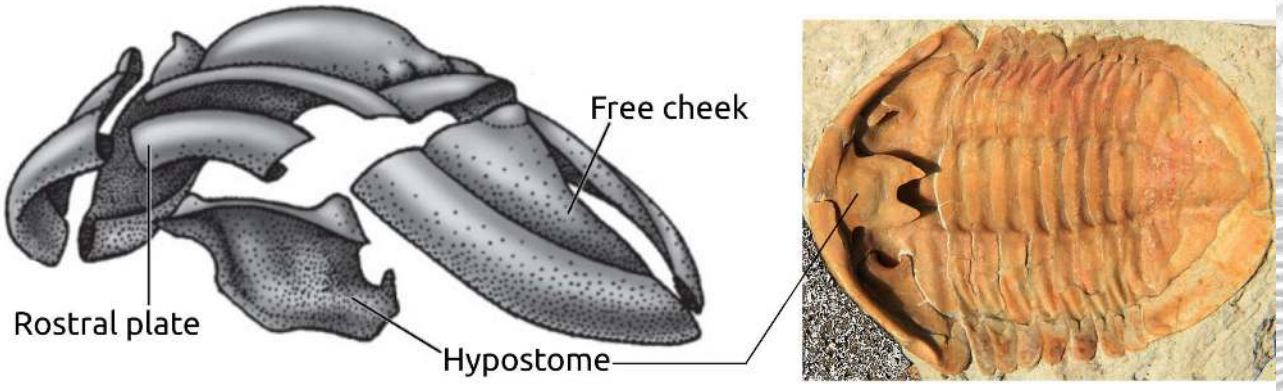
Axial/median lobe; Pleural/lateral lobe: All three regions were divided by a pair of furrows running the length of the body, giving a trilobite appearance (i.e. a median or axial lobe, flanked on either side by a lateral lobe).





Glabella: The raised median part of the cephalon which is very variable in size and structure and it is probable that in life the stomach lay beneath it.

Source right: Fortey, R. and Chatterton, B., 2003. A Devonian trilobite with an eyeshade. Science, 301(5640), 1689-1689.



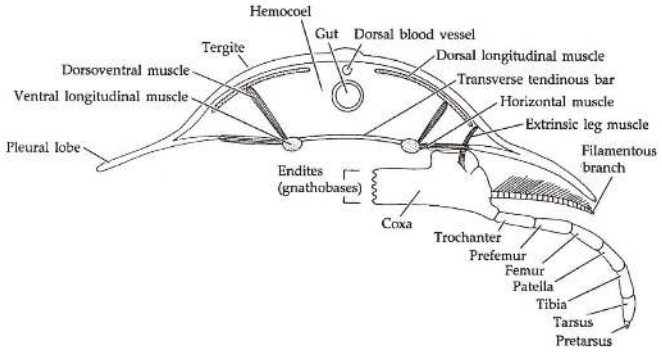
Rostral plate: A plate plate situated at the anterior margin of the head.

Hypostome: A plate posterior to the rostral plate, the hypostome, a plate of variable shape and size, is usually sited under the glabella. The shape and position of the hypostome is of great help in classifying the group.

Left: Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons.
Right: Dwergen Paartje.



Image from Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons.



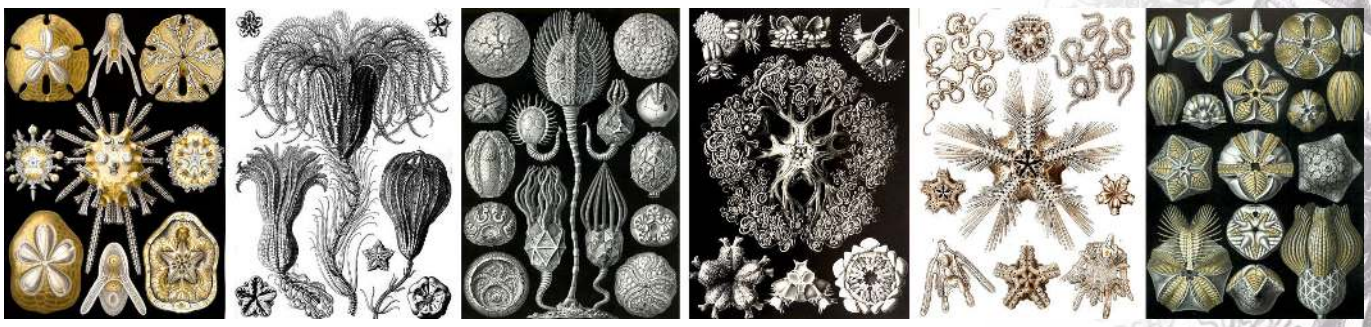


Left: Brusca, R.C. and Brusca, G.J., 2003. Invertebrates Sinnauer Associates.
Right: Copyright National Museum of Wales.



All images by Dwergen Paartje

EART27201 Sedimentary Rocks and Fossils



Macrofossils: Echinoderms

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Introduction and biology

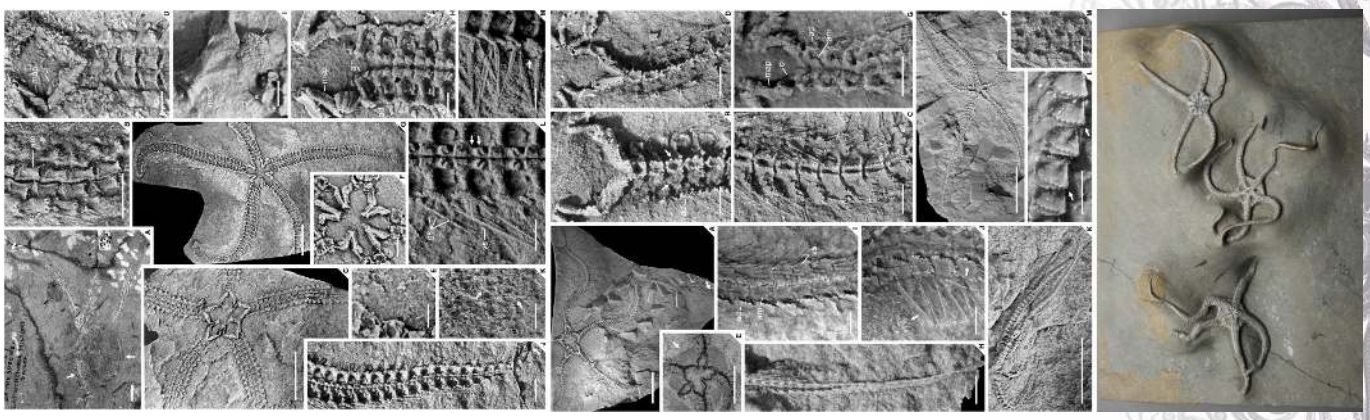
What is an echinoderm?



Stereom: In Echinodermata, a mesh, mainly of magnesian calcite, from which the skeleton is constructed.

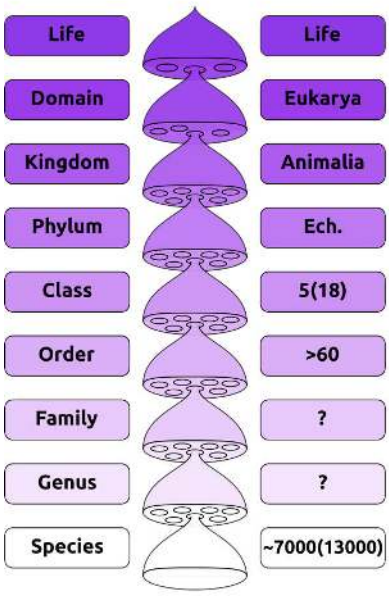
Top left: Brittle star - Florida Keys National Marine Sanctuary; Bottom left: sea cucumber, photographed by Anders Poulsen; Middle top: sea urchin, photograph by Kirt L. Onthank; Middle bottom: Red-knobbed Starfish by Adrian Pingstone. Definition: Allaby, M., 2013. A dictionary of geology and earth sciences. Oxford University Press.

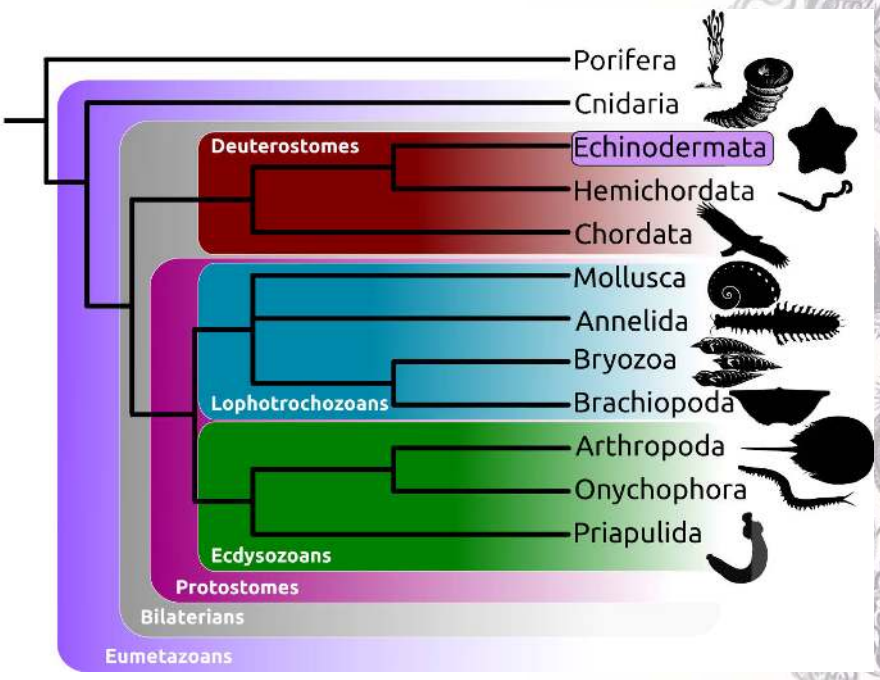
What are they like as fossils?

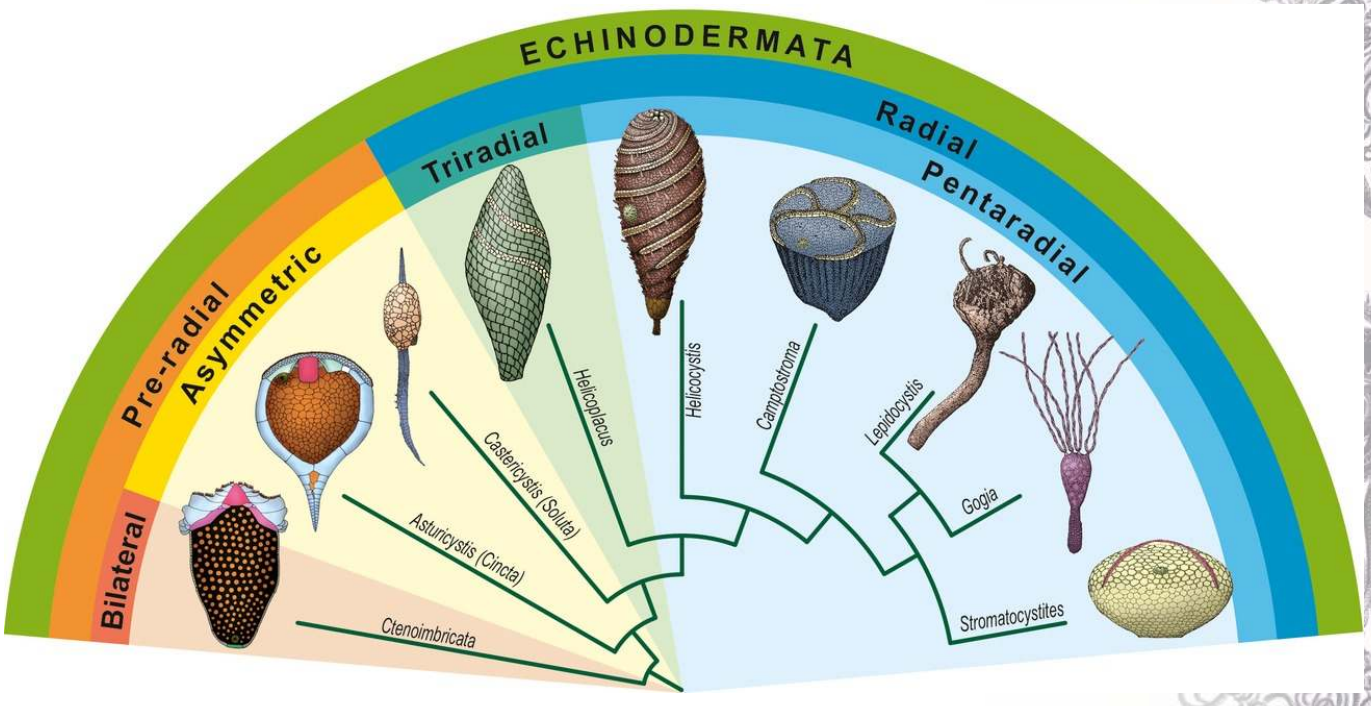


Left: Gladwell, D.J., 2018. Asterozoans from the Ludlow Series (upper Silurian) of Leintwardine, Herefordshire, UK. Papers in Palaeontology, 4(1), pp.101-160.
Right: © Jurassic Coast World Heritage Site Partnership.

Where do they sit on the tree of life?

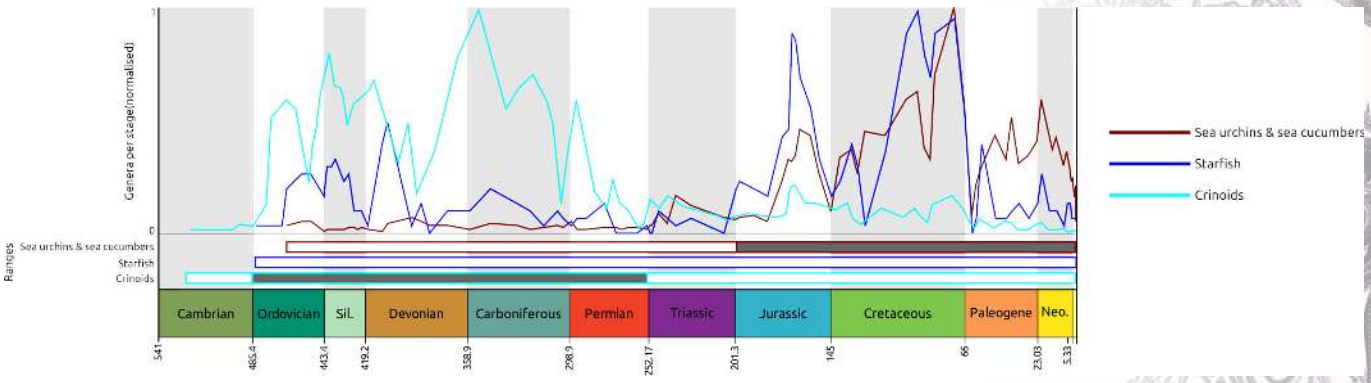






Zamora, S. and Rahman, I.A., 2014. Deciphering the early evolution of echinoderms with C ambrian fossils. Palaeontology, 57(6), pp.1105-1119.

How are they distributed through time?



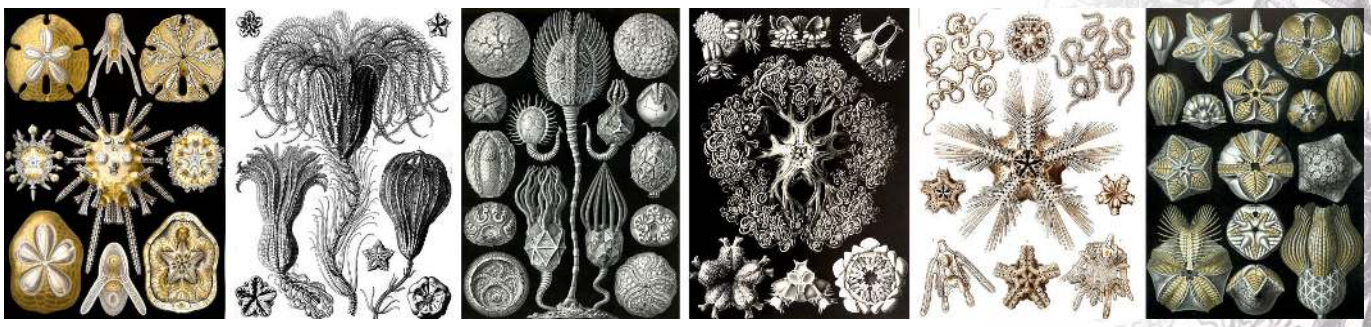
Fossil data from PBDB.

Why do they matter?



Finding nemo; Pixar.

EART27201 Sedimentary Rocks and Fossils



Macrofossils: Echinoderms

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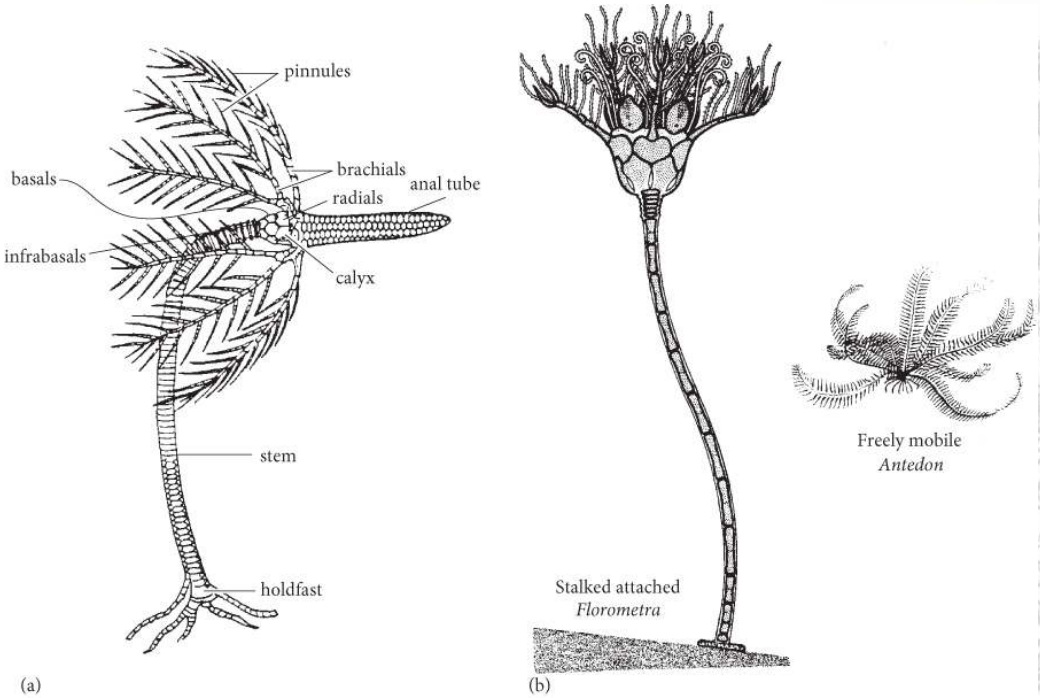
Morphology / Anatomy
Crinoidea



Crinoid: A living class of echinoderms, whose members are either stalked (sea lilies) or unstalked (feather stars).

Toarcian: A stage in the European Early Jurassic (183-175.6 Ma ago)

Photo by Jenna Davenport.



Source: Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons.

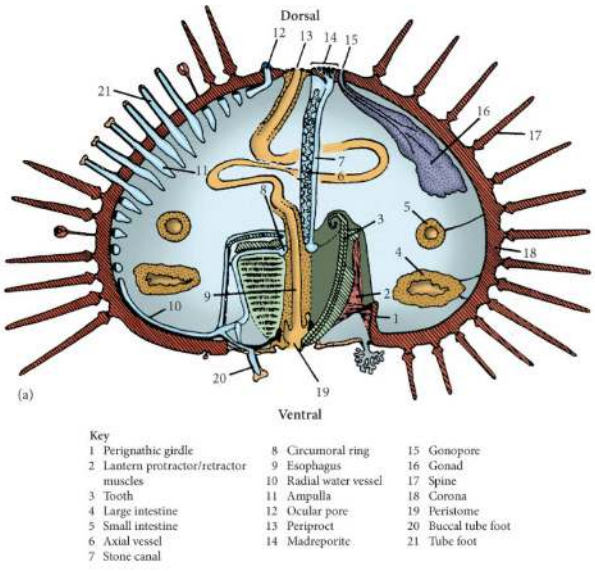
Ossicles: In Echinodermata, irregularly fenestrated calcareous plates, rods, and crosses arranged to form a lattice and bound together by connective tissue; together they comprise the skeleton.

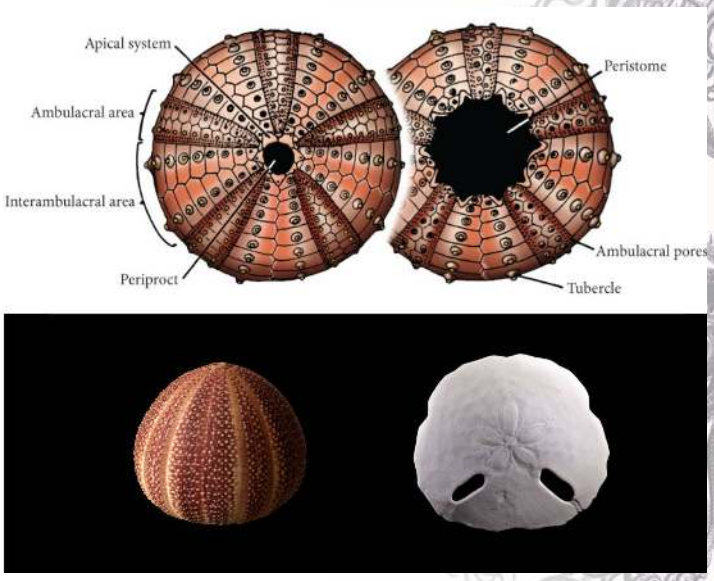
Holdfast: The rooting structure that fixes an archaeocyathan, a crinoid, a dendroid graptolite or a seaweed to a rock.

Calyx/aboral cup/theca: Cup-shaped, plated body of crinoids. The calyx is made up of several rows of plates.

Brachials (=arms): In crinoids (Crinoidea), long, plated, flexible arms made up of a series of articulated ossicles containing a water vessel from which the tube feet arise.

Morphology / Anatomy
Echinoidea





Source - Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons, except sand dollar, by Didier Descouens and regular sea urchin by Esko Jäntti.

Echinoid: A free-living echinoderm in which the body is enclosed in a globular, cushion-shaped, discoidal, or heart-shaped shell built of columns of interlocking, calcareous plates, which bear movable appendages.

Test: Protective shell that covers the cells of some protists and the soft parts of certain invertebrate animals.

Adapical/oral: In an echinoderm, the lower, adapical or oral, surface is perforated by the mouth.

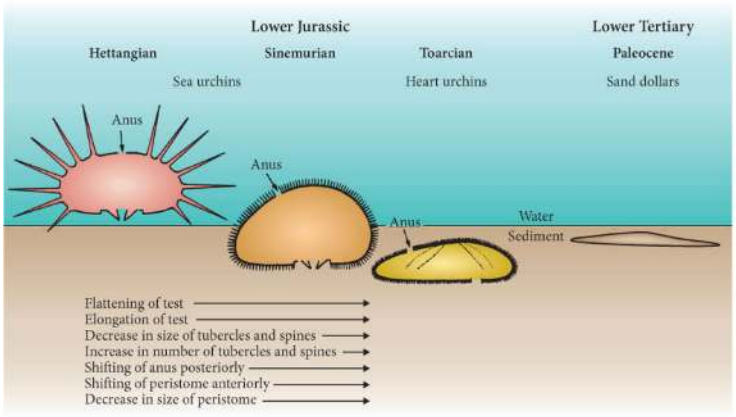
Apical/aboral: The upper, apical or aboral, surface that has the anal opening.

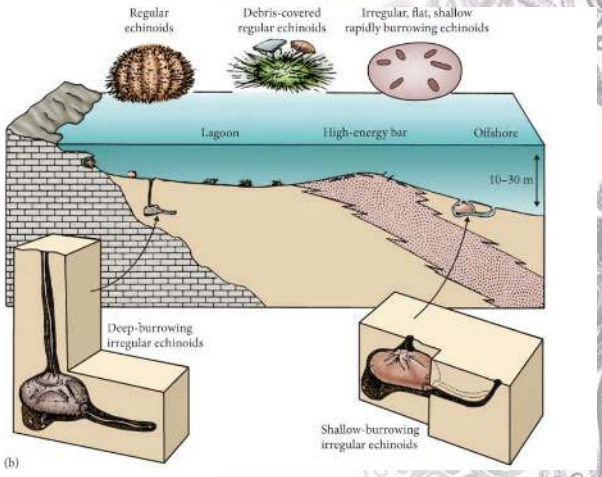
Ambulacrum: In Echinodermata, an area of the body surface (covered in most classes by calcitic plates), that overlies one of the radial canals of the internal water vascular system, and bears the tube feet.

Interambulacrum: In Echinodermata, that area of the body surface lying between ambulacra.

Periproct: In Echinoidea, the area surrounding the anus and covered in leathery skin in which small, calcitic plates are loosely embedded.

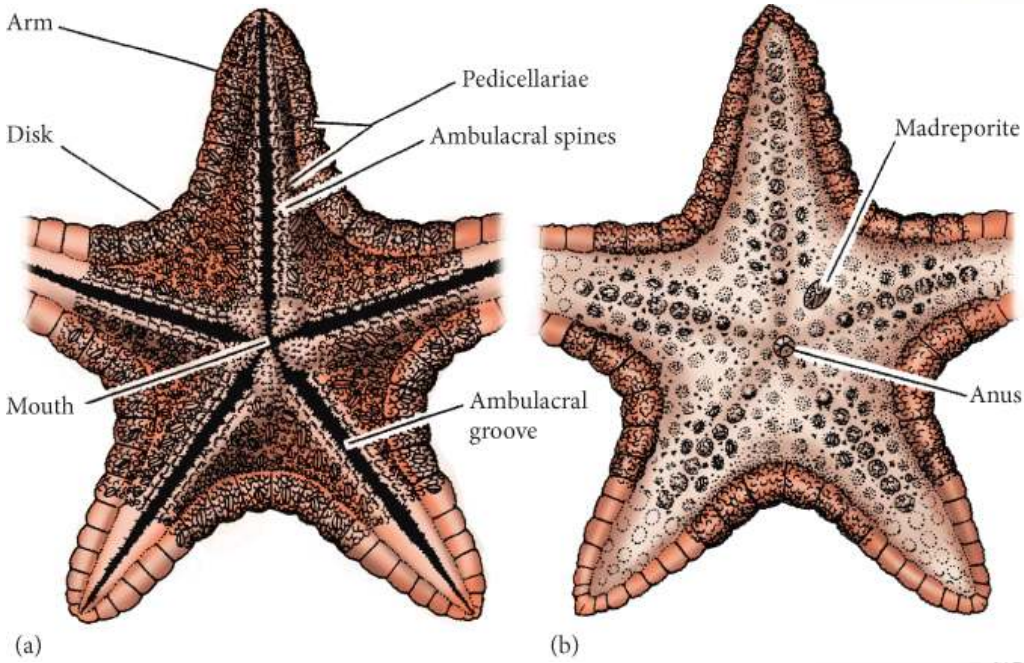
Peristome: In Echinoidea, the area around the mouth, covered in a leathery skin studded with small plates.





Source: Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons.

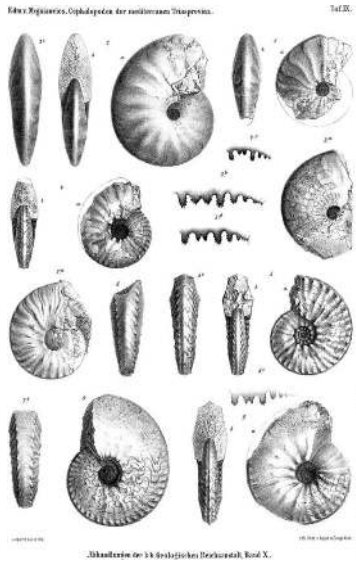
Morphology / Anatomy
Asteroidea and Ophiuroidea

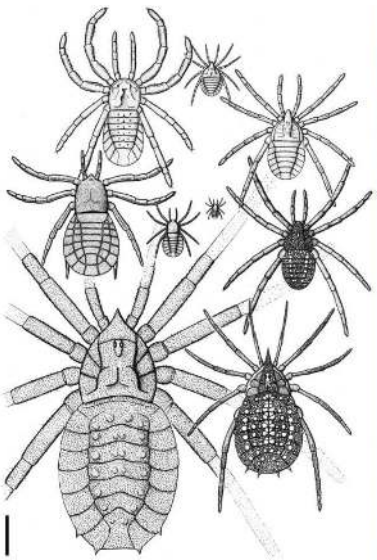


Asterozoa: Starfish (Asteroidea) and brittle stars (Ophiuroidea) — a subphylum whose members have radial symmetry of projecting rays, and a star-shaped body.

Source - Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons.

EART27201 Sedimentary Rocks and Fossils







Macrofossils: Biostratigraphy – 2

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Remember biozones?

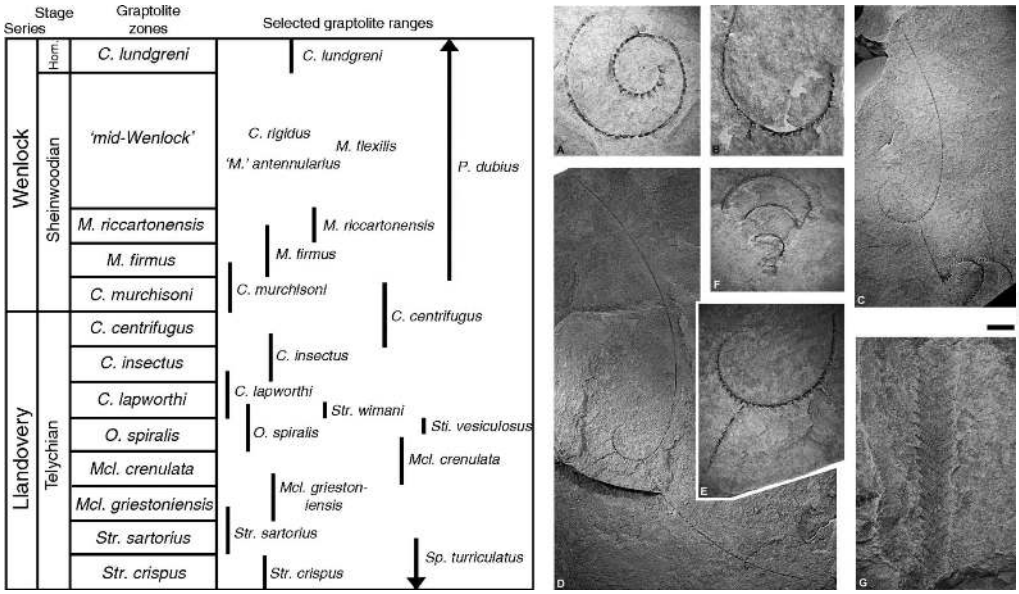
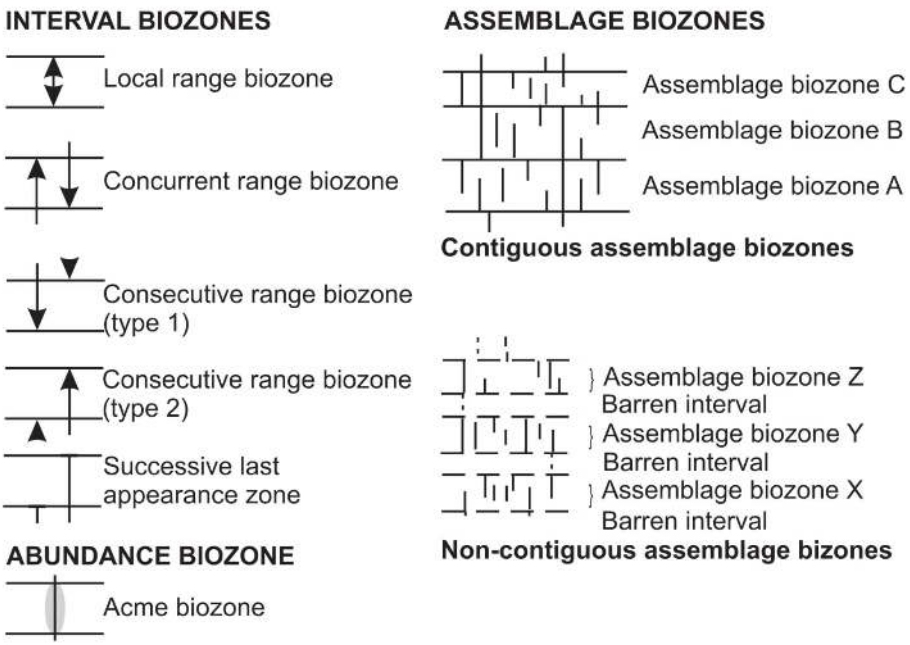


Image source: Cramer, B.D., et al. 2010. Testing the limits of Paleozoic chronostratigraphic correlation via high-resolution (< 500 ky) integrated conodont, graptolite, and carbon isotope (δ13Ccarb) biochemostratigraphy across the Llandovery–Wenlock (Silurian) boundary: Is a unified Phanerozoic time scale achievable? Geological Society of America Bulletin, 122(9-10), pp.1700-1716.
Arvestål, E.H. and Streng, M., 2013. Cyrtograptids from the Telychian (upper Llandovery) of Kinnekulle Mountain, southern Sweden. GFF, 135(1), pp.45-56.

Biostratigraphy - Biozone types



What's a FAD?

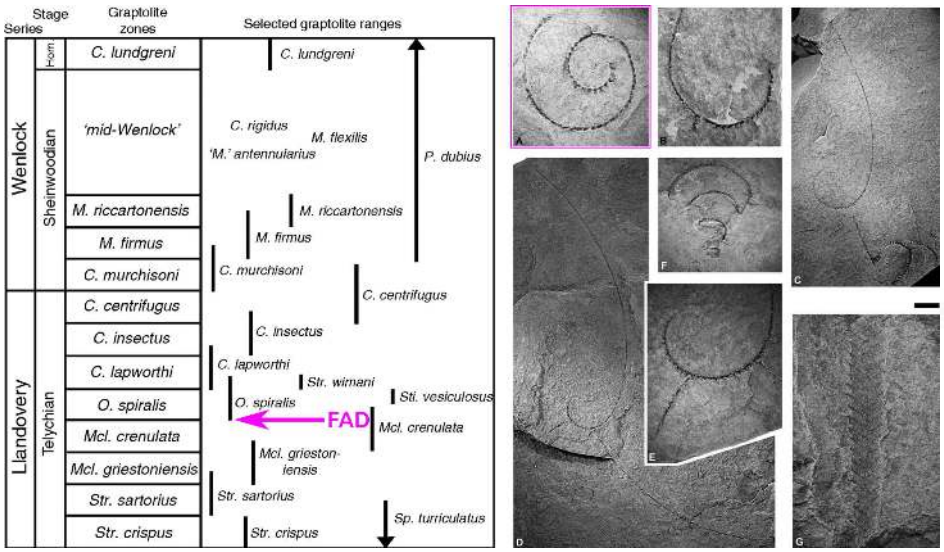


Image source: Cramer, B.D., et al. 2010. Testing the limits of Paleozoic chronostratigraphic correlation via high-resolution (< 500 ky) integrated conodont, graptolite, and carbon isotope (δ13Ccarb) biochemostratigraphy across the Llandovery-Wenlock (Silurian) boundary: Is a unified Phanerozoic time scale achievable? Geological Society of America Bulletin, 122(9-10), pp.1700-1716.
Arvestål, E.H. and Streng, M., 2013. Cyrtograptids from the Telychian (upper Llandovery) of Kinnekulle Mountain, southern Sweden. GFF, 135(1), pp.45-56.

First Appearance Datum (FAD)

And what, exactly, is a LAD?

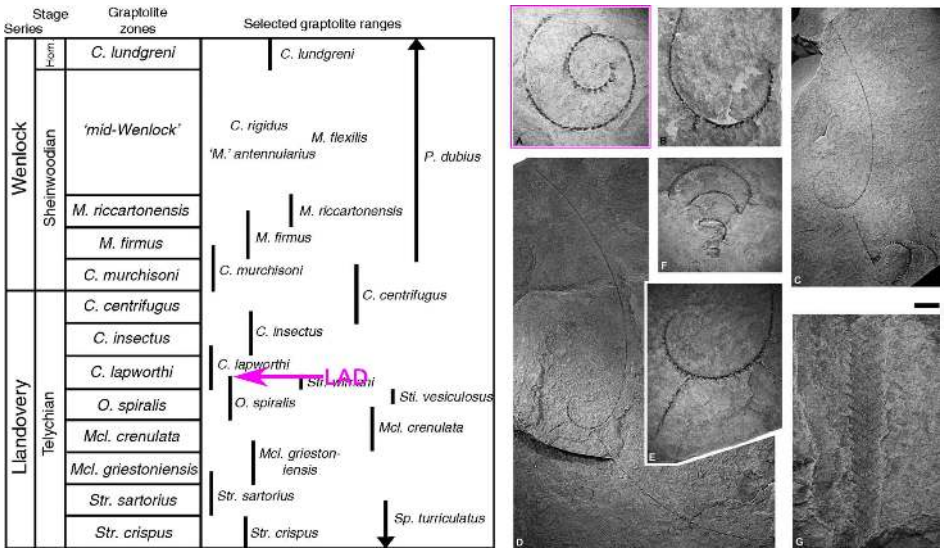


Image source: Cramer, B.D., et al. 2010. Testing the limits of Paleozoic chronostratigraphic correlation via high-resolution (< 500 ky) integrated conodont, graptolite, and carbon isotope (δ13Ccarb) biochemostratigraphy across the Llandovery–Wenlock (Silurian) boundary: Is a unified Phanerozoic time scale achievable? Geological Society of America Bulletin, 122(9-10), pp.1700-1716.
Arvestål, E.H. and Streng, M., 2013. Cyrtograptids from the Telychian (upper Llandovery) of Kinnekulle Mountain, southern Sweden. GFF, 135(1), pp.45-56.

Last Appearance Datum (LAD)

How do we present this data?

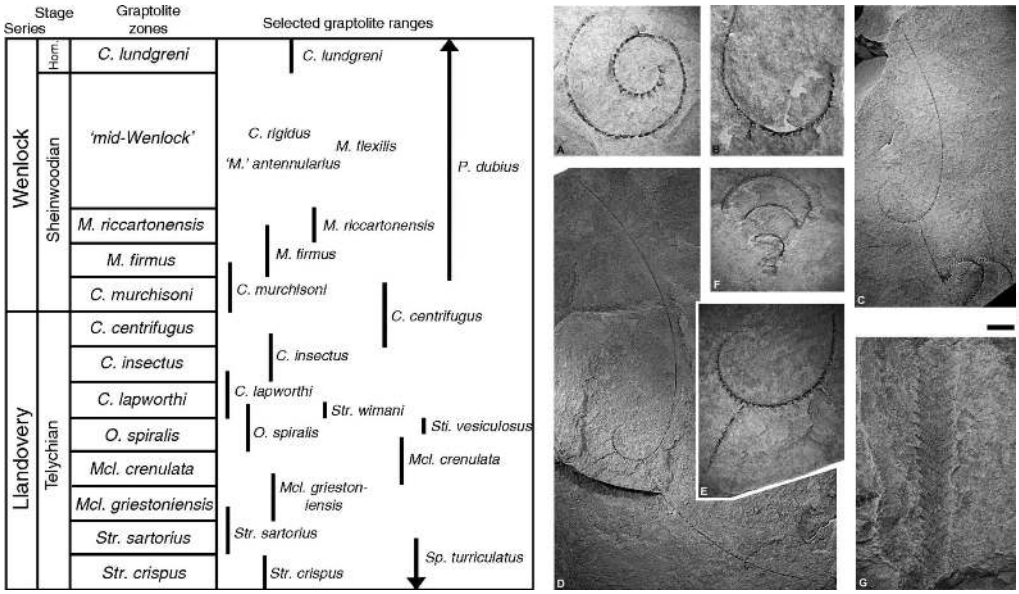
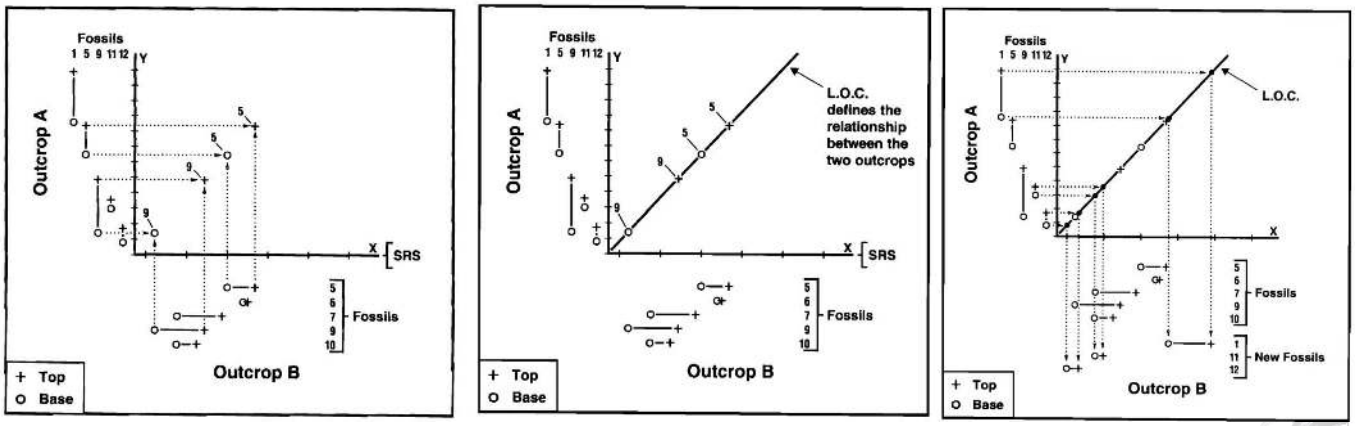


Image source: Cramer, B.D., et al. 2010. Testing the limits of Paleozoic chronostratigraphic correlation via high-resolution (< 500 ky) integrated conodont, graptolite, and carbon isotope (δ13Ccarb) biochemostratigraphy across the Llandovery–Wenlock (Silurian) boundary: Is a unified Phanerozoic time scale achievable? Geological Society of America Bulletin, 122(9-10), pp.1700-1716.

Arvestål, E.H. and Streng, M., 2013. Cyrtograptids from the Telychian (upper Llandovery) of Kinnekulle Mountain, southern Sweden. GFF, 135(1), pp.45-56.

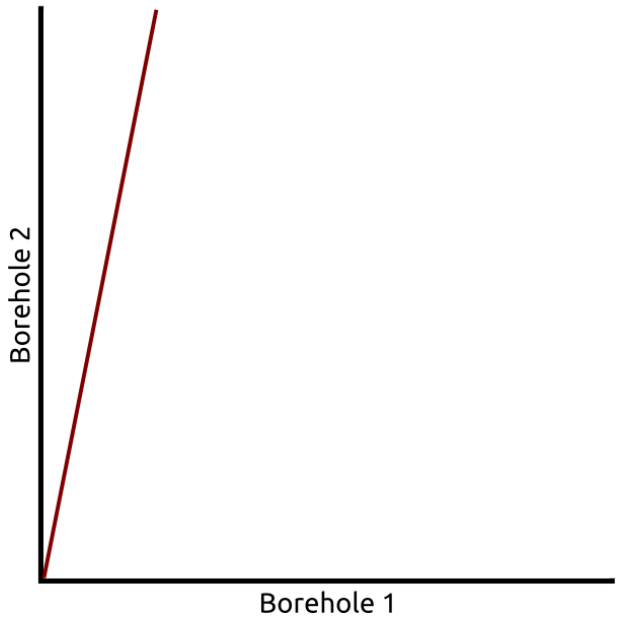
So what can we do with all of this?



Carney, J.L. and Pierce, R.W., 1995. Graphic correlation and composite standard databases as tools for the exploration biostratigrapher. In: Graphic Correlation Edited by Keith Olin Mann and H. Richard Lane. Vol. 53. 1995. P.23-43

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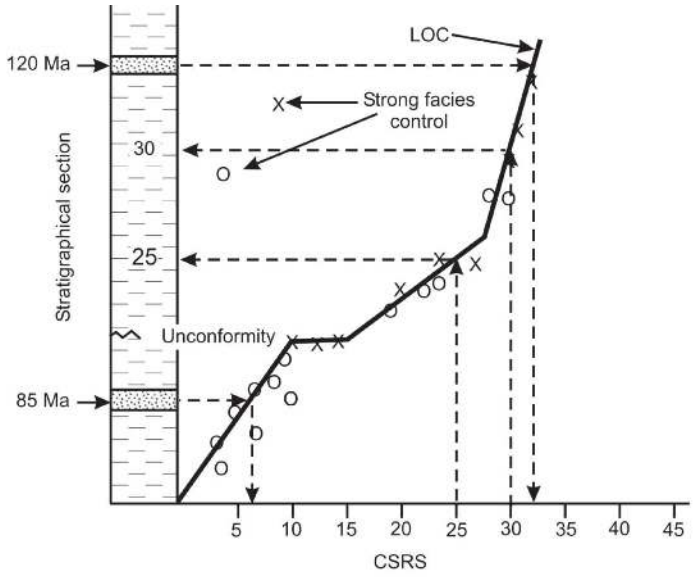
Which borehole has higher sedimentation rates?



What is the sedimentation history here?



Biostratigraphy - Correlation



CSRS = Composite standard reference section

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